

Project Proposal for Vizualizing HAT's GOE Monitoring Data

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Idea for Draft Project Proposal

- I am a student in UVic's Restoration of Natural Systems program interested in making a significant contribution to the local restoration community, and I'm looking to assist an interested community partner through the creation of a Final Restoration Project
- Nancy Shackelford (RNS Director) suggested that I might consider helping HAT GOE Monitoring Project by creating a tool for vizualizing GOE Monitoring Data.

- I propose to create a customized-for-HAT automatic coding tool that can be used by non-coders.
- I have already spent almost 30 hours (during August 2025) reviewing and learning the new-to-me R code I would need to complete this project, and feel confident I will be able to learn any other R code as will be deemed needed after consultation with HAT, and receiving HAT DATA to analyze its structure and formats

What I need

From HAT

- access to HAT GOE Monitoring Data
- support, input, and feedback in order
- I met with Vanessa Brownlee 2025-08-27, who asked that I send her further information about my proposed project, and though still new to her position, she seemed to be willing for me to start this project in Fall 2025 semester

From RNS

- support and supervision from RNS Nancy Shackelford, particularly before she goes on sabbatical July 2026

What I can offer

- A keen interest in restoring Garry Oak Ecosystems
- Extensive knowledge and experience with coding using R
- I am a self-motivated, low maintenance student, very interested in the natural environment, and who loves to continue learning in meaningful ways
- I can also assist HAT with the monitoring raw data clean-up as needed

Proposed Deliverables

Written report

- The report will provide a record of the work I did to complete this project
- Report will follow the format, length, and expectations of RNS program

Tutorials for using Data Visualization Tools

- easy to understand
- customized

RMarkdown Data Processing Document

- for data downloading, cleaning, and formatting data visualizations
- explain process for preparing data for use
- can be used to create HTML or pdf reports
- HTML Document includes a Table of Contents for easy navigation between parts

Shiny interactive tool

- <https://goe-interact.shinyapps.io/Test-Shiny-GOE-Monitor/>
- using data processed with RMarkdown document
- for choosing data variables to visualize in tables, plot charts, and maps

Vizualizations

- data tables, chart plots, maps

Data

- Data used for this proposal is from the report that HAT used as a model for their GOE Monitoring Project
- Malloff, J., & Shackelford, N. (2024). Feeling the Pulse: Monitoring methods and initial outcomes in oak meadow ecosystems. Restoration Futures Lab at the University of Victoria.

Data download and formatting

- Data (in excel or csv format) is uploaded
- Data is formatted, saved, and managed to be useful for creating visualizations
- The code has been hidden for this report, highlighting only the resulting plots

Data Visualization Examples

Data Table (Malloff & Shackelford, 2024)

Full Data Table

Subregion Site Land_Manager Main_Restoration_Type Restoration_Intensity Area_ha Lat Lng Proportion_of_native_species Cultural_species_richness Exotic_species Trampling Herbivory Composite_Index Year Percentage_ns

Head of Site Information

Subregion	Site	Land_Manager	Main_Restoration_Type	Restoration_Intensity	Area_ha	Lat	Ln
Gulf Islands	Anniversary Island	GINPR	herbivore reduction	high	4.39	48.82292	-123.182
Gulf Islands	AVNR	Saltspring conservancy	herbivore reduction	minimal	20.54	48.80351	-123.442
Saanich Peninsula	Bear Hill Park	CRD	invasive removal	low	3.80	48.54639	-123.407
Gulf Islands	Brackman Island	GINPR	invasive removal	high	4.41	48.71897	-123.386
Saanich Peninsula	Camas Hill	HAT	invasive removal	high	10.10	48.40173	-123.597
Gulf Islands	Crows Nest	Trinity Western University	herbivore reduction	high	15.34	48.78237	-123.461

Head of Site Data

Subregion	Site	Proportion_of_native_species	Cultural_species_richness	Exotic_species	Trampling	Herbivory	Compos
Gulf Islands	Anniversary Island	0.94	2.71	0.17	5.00	1.14	
Gulf Islands	AVNR	0.55	1.30	57.19	4.70	36.50	
Saanich Peninsula	Bear Hill Park	0.62	1.86	24.46	7.57	9.50	
Gulf Islands	Brackman Island	0.77	2.80	11.17	5.40	0.00	
Saanich Peninsula	Camas Hill	0.61	2.43	27.19	7.71	4.71	
Gulf Islands	Crows Nest	0.66	1.07	31.51	1.64	26.71	

Map of Site Locations

Leaflet Map with Circle Markers

- Data from previous data table is summarized in the map marker popups
- click markers to open site data summary window

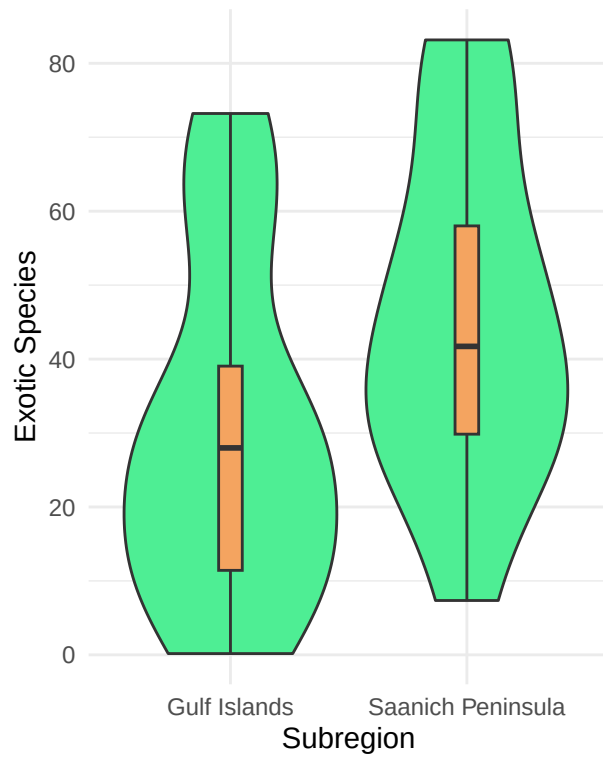


Subregion Plots

Violin/Bar Plot: Compare Subregions Exotic and Native Species

A Exotic Species

ECCC (Malloff & Shackelford, 2024)



B Native Species

ECCC (Malloff & Shackelford, 2024)

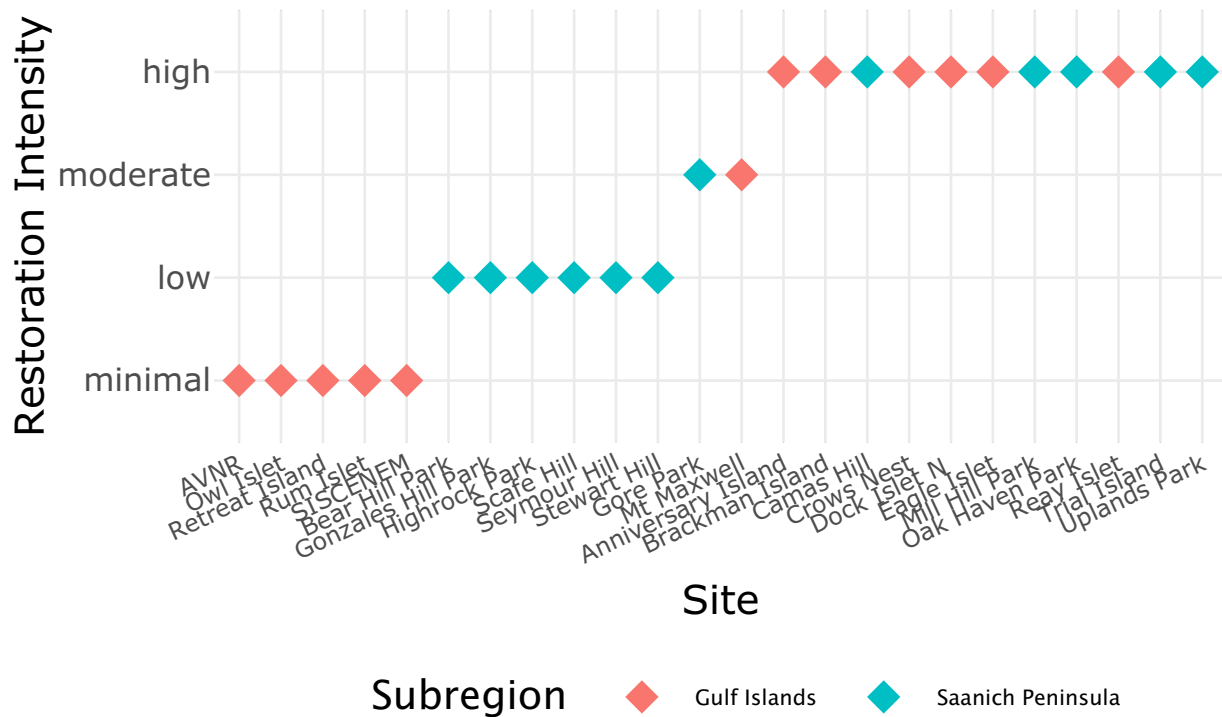


Chart by Wendy Anthony
2025-08-27

Point Plot: Sites Ordered by Restoration Intensity

- this uses the same plot code as the previous example, changing only the order of the restoration intensity from minimal to high
- click data points for tool-tip of data results

Plot Site Restoration Intensity



Point Plot: Individual Site - Uplands Variables of Degradation

Plotting Uplands Degradation Variables

ECCC (Malloff & Shackelford, 2024)

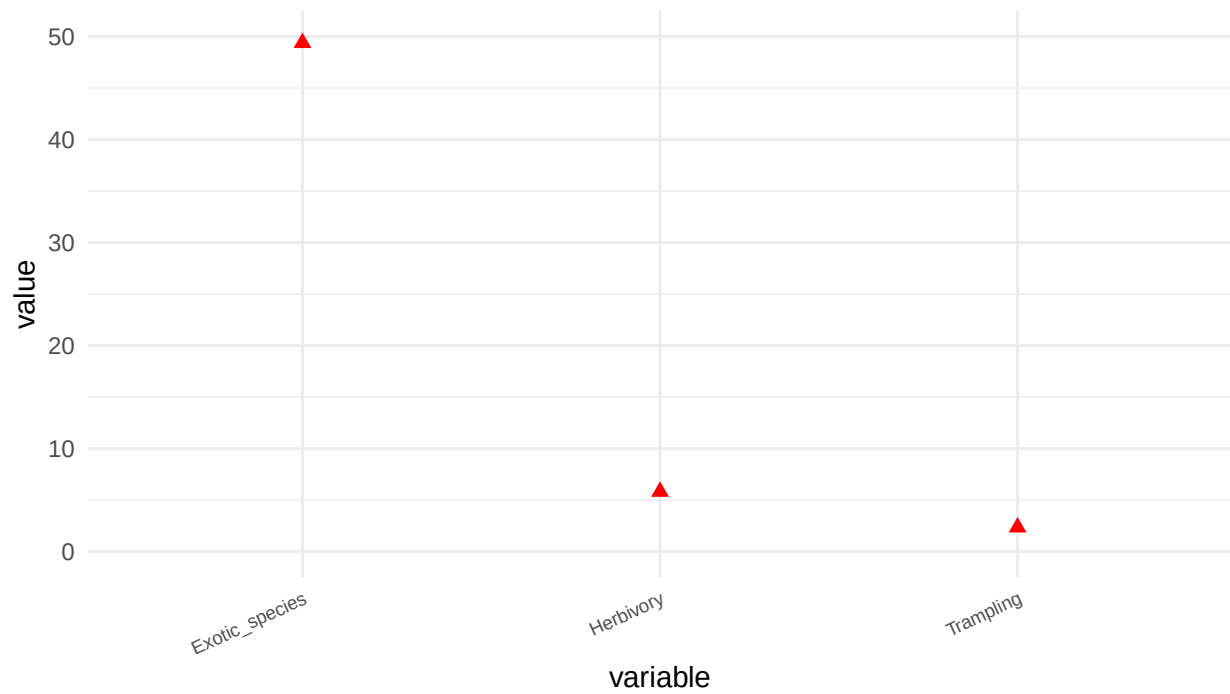


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